

Bayesian Estimation of Human Population Toxicokinetics of PFOA, PFOS, PFHxS, and PFNA from Studies of Contaminated Drinking Water

Weihsueh A. Chiu,^{1,2} Meghan T. Lynch,³ Claire R. Lay,³ Adriana Antezana,³ Parker Malek,³ Sara Sokolinski,³ and Rachel D. Rogers⁴

¹Interdisciplinary Faculty of Toxicology, Texas A&M University, College Station, Texas, USA

²Department of Veterinary Physiology and Pharmacology, School of Veterinary Medicine and Biomedical Sciences, Texas A&M University, College Station, Texas, USA

³Abt Associates, Cambridge, Massachusetts, USA

⁴Centers for Disease Control and Prevention/Agency for Toxic Substances and Disease Registry, Atlanta, Georgia, USA

BACKGROUND: Setting health-protective standards for poly- and perfluoroalkyl substances (PFAS) exposure requires estimates of their population toxicokinetics, but existing studies have reported widely varying PFAS half-lives ($T_{1/2}$) and volumes of distribution (V_d).

OBJECTIVES: We combined data from multiple studies to develop harmonized estimates of $T_{1/2}$ and V_d , along with their interindividual variability, for four PFAS commonly found in drinking water: perfluorooctanoic acid (PFOA), perfluorooctane sulfonate (PFOS), perfluorononanoic acid (PFNA), and perfluorohexane sulfonate (PFHxS).

METHODS: We identified published data on PFAS concentrations in human serum with corresponding drinking water measurements, separated into training and testing data sets. We fit training data sets to a one-compartment model incorporating interindividual variability, time-dependent drinking water concentrations, and background exposures. Use of a hierarchical Bayesian approach allowed us to incorporate informative priors at the population level, as well as at the study level. We compared posterior predictions to testing data sets to evaluate model performance.

RESULTS: Posterior median (95% CI) estimates of $T_{1/2}$ (in years) for the population geometric mean were 3.14 (2.69, 3.73) for PFOA, 3.36 (2.52, 4.42) for PFOS, 2.35 (1.65, 3.16) for PFNA, and 8.30 (5.38, 13.5) for PFHxS, all of which were within the range of previously published values. The extensive individual-level data for PFOA allowed accurate estimation of population variability, with a population geometric standard deviation of 1.57 (95% CI: 1.42, 1.73); data from other PFAS were also consistent with this degree of population variability. V_d estimates ranged from 0.19 to 0.43 L/kg across the four PFAS, which tended to be slightly higher than previously published estimates.

DISCUSSION: These results have direct application in both risk assessment (quantitative interspecies extrapolation and uncertainty factors for interindividual variability) and risk communication (interpretation of monitoring data). In addition, this study provides a rigorous methodology for further refinement with additional data, as well as application to other PFAS. <https://doi.org/10.1289/EHP10103>

Introduction

Poly- and perfluoroalkyl substances (PFAS) are man-made chemicals consisting of chains of linked carbon and fluorine atoms. Their structure gives them unique physicochemical properties that have been found useful in a range of consumer and industrial products. However, these properties also lead to high water solubility, bioaccumulation, persistence in the environment, and resistance to degradation, leading them to be called “forever chemicals.” Owing to the widespread use of PFAS and their high solubility in water, PFAS contamination has been reported in drinking water throughout the United States (Cordner et al. 2019; Evans et al. 2020; Hu et al. 2016). Biomonitoring data have confirmed that PFAS exposure is widespread in humans (CDC 2019; Daly et al. 2018; Olsen et al. 2017; Yu et al. 2020). The National Health and Nutrition Examination Survey (NHANES) has measured PFAS in serum samples from the general population since 1999, finding detectable levels in the blood of 99% of the population >12 years of age (CDC 2019). Finally, several well-studied PFAS, including perfluorooctanoic acid (PFOA) and perfluorooctane

sulfonic acid (PFOS), have been linked to adverse human health effects (ATSDR 2021; IARC 2017; NTP 2016; Sunderland et al. 2019).

For the many thousands of other PFAS that have been released into the environment, few toxicological data exist. Some PFAS have been phased out in manufacturing processes in much of the world, and serum concentrations for some long-chain PFAS are declining in the United States (CDC 2019). However, PFAS contamination in water persists, and tools to estimate exposure and health effects from drinking water concentrations (DWCs) are useful for public health and regulatory work. This study used data from published human studies linking DWCs to serum concentrations to produce robust estimates of key pharmacokinetic parameters for four PFAS.

Since the early 2000s, the U.S. Environmental Protection Agency (EPA) and many state agencies have embarked on monitoring programs and the development of drinking water advisory levels (e.g., CA Water Boards 2020; MDH 2021; EGLE 2020; NJDEP 2021; U.S. EPA 2016). State guidelines are generally <50 ng/L. For instance, the Minnesota Department of Health (MDH) has set drinking water standards for the state for perfluorohexane sulfonate (PFHxS), PFOA, and PFOS of 47, 35, and 15 ng/L, respectively (MDH 2021). The State of New Jersey has established enforceable drinking water standards for PFOA, PFOS, and perfluorononanoic acid (PFNA) where the maximum contaminant levels are set at 14, 13, and 13 ng/L, respectively (NJDEP 2021). The State of California has established notification levels for PFOS and PFOA of 6.5 and 5.1 ng/L, respectively (CA Water Boards 2020). The State of Michigan has set maximum contaminant levels for PFNA, PFOA, PFOS, and PFHxS at 6, 8, 16, and 51 ng/L, respectively (EGLE 2020). Most recently, the U.S. EPA has set much lower interim health advisory levels for PFOA (0.004 ng/L) and PFOS (0.02 ng/L) (U.S. EPA 2022).

Address correspondence to Meghan T. Lynch, 10 Fawcett St., Cambridge, MA 02138 USA. Email: mtklynch@gmail.com

Supplemental Material is available online (<https://doi.org/10.1289/EHP10103>).

The authors declare they have no actual or potential competing financial interests.

Received 9 August 2021; Revised 3 August 2022; Accepted 27 October 2022; Published 1 December 2022.

Note to readers with disabilities: EHP strives to ensure that all journal content is accessible to all readers. However, some figures and Supplemental Material published in EHP articles may not conform to 508 standards due to the complexity of the information being presented. If you need assistance accessing journal content, please contact ehpsubmissions@niehs.nih.gov. Our staff will work with you to assess and meet your accessibility needs within 3 working days.

A critical component for risk assessment and risk communication regarding PFAS involves quantifying their toxicokinetics in human populations. Several widespread PFAS appear to be eliminated extremely slowly in humans, with half-life ($T_{1/2}$, the time it takes for serum concentrations to decline by 50%) measured in years, whereas $T_{1/2}$ in experimental animals are on the order of days or weeks. Numerous human toxicokinetics models for individual PFAS have been developed, ranging from simple compartmental models to physiologically based pharmacokinetic (PBPK) models. However, for regulatory and public health applications, the focus has been on the one-compartment model (Egeghy and Lorber 2011; Lorber and Egeghy 2011). The key parameters for this one-compartment model are the $T_{1/2}$ and volume of distribution (V_d , representing a chemical's tendency to remain in the blood or to distribute to other body tissues), given that these parameters together are sufficient to derive chemical-specific adjustment factors for interspecies and interindividual variability (WHO et al. 2005; U.S. EPA 2014). However, existing human studies report wide ranges of $T_{1/2}$ even for relatively well-studied PFAS. For example, depending on the study and population, $T_{1/2}$ estimates for PFOA have ranged from 3 to 10 y, and estimates for PFOS range even wider, from 3 to 27 y (Bartell et al. 2010; Costa et al. 2009; Harada et al. 2005; Li et al. 2018; Olsen et al. 2007; Seals et al. 2011; Wong et al. 2014; Worley et al. 2017a, 2017b; Zhang et al. 2013). Several studies have also reported substantial interindividual variability in $T_{1/2}$ (Li et al. 2018; Olsen et al. 2007). Moreover, the data available on V_d are sparse compared with the data available for $T_{1/2}$ (Koponen et al. 2018). Accurate estimates of V_d are important for converting between $T_{1/2}$ and clearance (e.g., Zhang et al. 2013).

Estimating V_d and $T_{1/2}$ for long-lived substances such as PFAS is challenging for several reasons. First, time-course data are required over several years to accurately estimate elimination. However, the ubiquity of PFAS means that ongoing exposure needs to be well characterized to avoid overestimating $T_{1/2}$. For V_d it is also necessary to have quantitative information on levels of exposure, which is often unavailable. Further, interindividual variation can make it difficult to distinguish measurement error from true heterogeneity. Characterizing this variation is also important to ensure adequate health protection across the population, including susceptible subgroups. Given that V_d and $T_{1/2}$ are the main parameters in one-compartment pharmacokinetic models, including those that include subcompartments addressing placental or lactational transfer, harmonized estimates of their values, along with estimates of the extent of interindividual variability, are needed to support public health actions related to PFAS.

In this study, we aimed to estimate the population toxicokinetics of four PFAS often found in drinking water: PFOA, PFOS, PFNA, and PFHxS (U.S. EPA 2017; EWG 2019a). Unlike previous studies that analyzed individual data sets, we took a more integrative approach and combined individual-level data from multiple studies for which detailed information on drinking water contamination, background exposures, and serum concentrations were available. Similar to previous work integrating PFAS toxicokinetics data across nonhuman animal species (Wambaugh et al. 2013), we employed a Bayesian approach, which enabled incorporation of prior knowledge, statistically rigorous incorporation of multiple data sets, better accommodation of unobserved variables, and quantitative characterization of uncertainty (see Dunson 2001; Silver 2012; Nuzzo 2015). Although this work is complementary to Wambaugh et al. (2013), because of the importance of characterizing human interindividual variability, we used a hierarchical Bayesian approach that adds random effects to model population variability.

Methods

Water and Serum PFAS Data

The goal of this data collection was to find published PFAS concentrations in human serum and drinking water levels to predict those serum levels. We located human serum concentrations and corresponding data on DWCs by searching databases such as PubMed, ScienceDirect, and Google Scholar for English language journal articles and reports that presented human serum data for PFOA, PFOS, PFNA, or PFHxS levels. We also conducted a tree search on identified review articles and reports. We did not apply constraints on publication date, but the literature review to identify studies was concluded in 2019. We focused on populations exposed through contaminated drinking water, as defined by study authors. When serum data for additional PFAS were identified in studies where contamination of one PFAS was indicated prior to conducting the study, we did also collect these serum concentrations and include them if we could also identify corresponding drinking water information. In some cases, when PFAS levels in drinking water were reported as below the minimum reporting level (MRL) for the U.S. EPA's third Unregulated Contaminant Monitoring Rule (UCMR3), but identified in serum of the study participants, we estimated the DWC as below the MRL as part of the analysis. We excluded studies focused on occupational cohorts, and where possible, we excluded individuals in community studies also known to have likely occupational exposures, such as employees of some chemical companies or firefighters, because our goal was to develop a tool that is predictive of community PFAS levels. For individual-level data, this information was only available for the Decatur, Alabama, data set, and so other data sets may include individuals with significant occupational exposure. Although several studies collected some information about occupation, only Bartell et al. (2010) and Emmett et al. (2006) specifically detailed how they excluded those with known or likely occupational exposure to the PFAS. All data were obtained from previous publications, and this analysis did not require institutional review board approval.

To be included in our model, a study needed to provide sufficient detail so that we could understand the timing of the PFAS exposure and map the serum levels to relevant DWCs for the study participants if the study did not explicitly give these concentrations. We reviewed the background, methods, results, figures, and tables of each study for reporting of community- and individual-level human serum data. Data extraction included recording the individual serum level if available, as well as the mean, geometric mean (GM), median, minimum, maximum, standard deviation (SD), sample size, dates of sampling, geographic location, and relevant demographic information, such as participant age and sex. When they were provided, we recorded dates of PFAS water contamination and remediation to better estimate participant exposure. If a study provided water consumption or finished water concentration data along with serum levels, we also collected this information. We aimed to develop as complete a picture as possible of DWCs over time prior to the serum measurements. When participant geographic information such as ZIP code, city, or water district service area was reported, we searched PFAS water concentration data to match them to serum data by location and dates. We obtained finished water concentration data for matching locations from journal articles, service districts' water quality reports, and data collected under the U.S. EPA's UCMR3. We used the Environmental Working Group's (EWG's) National Tap Water Database (U.S. EPA 2017; EWG 2019a) to identify additional sources of drinking water concentration data but verified data contained in the EWG database with the original sources. We did not include source water concentrations, and we included finished

water data for all available time periods, including during participant sampling, before remediation, or after remediation, if available. Data recorded included water collection data, water type, arithmetic mean, GM, median, minimum, maximum, SD, and number of samples. The authors of several human serum studies reported the PFAS concentration that participants were exposed to in their drinking water, and in those cases, we recorded the PFAS concentration for the location directly from the serum-level studies.

To be included for analysis, studies with time-course serum data had to have available water concentration data during the time period between the first and last serum collection. For studies with only a single serum collection, water concentration data had to precede the serum collection date. If we could not reliably match study participants with reliable water concentrations, we excluded those serum levels. In addition, owing to limited data, no data sets focusing exclusively on children were included, and for a single study with both children and adults >18 years of age, only the adult values were used in estimation. We excluded individuals with a known potential for occupational PFAS exposure, those who could not be linked to specific water concentrations, those who reported filtering their water, and those with missing values for important variables such as weight and sex. We did include populations drinking bottled water in the testing data sets. We ultimately included data from nine sites for parameter estimation. Study data sets, citations, and location-specific water concentration handling are listed in Table 1 and, in further detail, in Tables S1 and S2. Table S1 includes details on why a location was excluded, which was typically due to a lack of sufficient information on water concentrations or serum concentrations.

Toxicokinetic Model

In keeping with previous and current PBPK modeling by the U.S. EPA in adult (nonpregnant) humans (Lorber and Egeghy 2011; U.S. EPA 2016), we used a one-compartment toxicokinetic model for each PFAS. Moreover, the parameters of the one-compartment model are also included in human PBPK models that include pregnancy and lactation (MDH 2021; NJDEP 2021). Thus, this approach will enable direct use of our modeling results in a wide range of PFAS public health assessments. The parameters were defined as follows: C_{bgd} is the background serum concentration (in micrograms per liter); C_0 is the initial serum concentration (in micrograms per liter) at $t=0$; DWI_{BW} is the drinking water intake on a body weight-adjusted basis (e.g., as liters per kilogram per day); $k = \ln(2)/T_{1/2}$ is the rate constant per day; and V_d is the V_d (in liters per kilogram).

All of these parameters are assumed to be constant for the duration of the simulation, except for the DWC, $DWC(t)$ (in micrograms per liter), which may vary over time. The ordinary differential equation (ODE) for the concentration of PFAS in the body is as follows:

$$\frac{dC(t)}{dt} = \frac{DWI_{BW}DWC(t)}{V_d} + k[C_{bgd} - C(t)]. \quad (1)$$

As shown in Figure 1, this model is equivalent to the usual one-compartment model used for pharmaceuticals, but with the usual bolus dose (typically modeled as either an initial condition or an exponential input) replaced by a time-varying input that includes both a drinking water component (first term) and a background exposure component. Moreover, the background dose rate (e.g., D_{bgd} , in micrograms per kilogram per day) has been reparameterized in terms of the background serum concentration, $C_{bgd} = D_{bgd}/(kV_d)$. This reparameterization improves model fitting by reducing parameter correlations. In addition,

because clearance is used as the basis for both chemical-specific interspecies extrapolation and interindividual variability factors (WHO et al. 2005; U.S. EPA 2014), CL (in liters per kilogram per day) $= k \times V_d$ was also calculated.

For a constant water concentration, Equation 2 has an analytic solution for the serum concentration as a function of time:

$$C(t) = C_{bgd} + (C_0 - C_{bgd})e^{-kt} + C_{ss}(1 - e^{-kt}), \quad (2)$$

where C_0 is the concentration at time 0, and C_{ss} , the steady-state serum concentration (due to drinking water alone, without background), is given by Equation 3:

$$C_{ss} = \frac{DWI_{BW}DWC}{kV_d}. \quad (3)$$

The first term in Equation 2 is simply the background concentration, the second term describes the time-dependence from time 0 to background in the absence of water contamination, and the third term describes the time-dependence of the transition to steady state. If data are available only at a single time point, then steady state is assumed, and DWCs are assumed to be constant:

$$C_{bgd+ss} = C_{bgd} + C_{ss}. \quad (4)$$

In some cases, the data at a single time point involve an intervention in which the source of contamination was removed at a certain time interval, Δt , in the past. This simplification of intervention is necessary when no water concentrations for after intervention are available. In this case, the serum concentration takes the following form:

$$C_{bgd+ss+\Delta t} = C_{bgd} + C_{ss}e^{-k\Delta t}. \quad (5)$$

Bayesian Population and Statistical Model

We employed a Bayesian population approach to estimate model parameters. This approach involves two statistical models: one for population variability and one for the likelihood of the observed data (serum concentrations) given a set of model parameters (such as $T_{1/2}$ and V_d). With respect to the population variability model, each individual i is assumed to have their own (unknown) model parameters θ_i and serum concentration data C_i at time t_j . The individual parameters (log-transformed) are drawn from a normal population distribution:

$$\log \theta_i = \mu_0 + \alpha_{\theta,i}\Sigma_0; \alpha_{\theta,i} \sim N(0,1). \quad (6)$$

Here, $\alpha_{\theta,i}$ represents the z-score of the individual in the population for parameter θ . For some parameters, the population mean, μ_0 , and SD, Σ_0 , may be fixed (e.g., background, drinking water intake), whereas for others, μ_0 and Σ_0 may have distributions that are due to uncertainty (e.g., $T_{1/2}$, V_d).

With respect to the statistical model for the likelihood of the observed data, we assume lognormal errors with an additional parameter to specify the SD of the error distribution, so the likelihood function for an observation, $C_{i,obs}(t_j)$, in individual i at time point t_j would be defined as follows:

$$C_{i,obs}(t_j) = C_{i,pred}(t_j)e^{\epsilon_{ij}}; \epsilon_{ij} \sim N(0, \sigma_{err}); GSD_{err} = e^{\sigma_{err}}, \quad (7)$$

where C_{pred} is the model prediction for serum concentration as a function of time as described above. Different values for the error of the geometric standard deviation (GSD_{err}) are used for individual time-course data, individual steady-state data, and summary-level steady-state data.

Table 1. Studies of PFAS-contaminated community drinking water and serum concentrations used.

City, state/country	Studies/data sources	Water data dates	Serum data dates (time points) (n)	PFAS data used	Individuals or (populations) (n)	Train/test data	LOD	Comments
Arnsberg, Germany	Hölzer et al. 2008, 2009	2006–2007	2006–2007 (2)	PFOA	220	110/110	10 ng/L	Time-varying water concentration between serum collections. PFOS only has summary data, but inadequate to calculate population arithmetic mean.
Decatur, Alabama	ATSDR 2013, 2016	2010–2016	2010–2016 (2)	PFOA, PFOS, PFNA, PFHxS	37	18/19	10 ng/L	Time-varying water concentration between serum collections.
Horsham, Pennsylvania	U.S. EPA 2017; HWSA 2014, 2018; Penn DOH 2019	2014–2015	2018 (1)	PFOA, PFOS, PFNA, PFHxS	(1)	(1/0)	20 ng/L (PFOA; PFNA) 30 ng/L (PFHxS) 40 ng/L (PFOS)	Assumed steady state until intervention in July 2016, 2 y prior to serum data collection.
Lake Elmo/ Cottage Grove, Minnesota	Johnson et al. 2017	2005–2008	2008 (1)	PFOA, PFOS	PFOA: 95, PFOS: 98	PFOA: 48/47; PFOS: 49/49	NR	Matched well water and serum measurements; assumed steady state.
Little Hocking, Ohio	Bartell et al. 2010	2007–2008	2007–2008 (2)	PFOA	(2)	(1/1)	16 ng/L	Public water population used for training, bottled water population for testing; only included data after intervention in November 2007.
Little Hocking, Ohio Lubeck, West Virginia	Emmett et al. 2006 Bartell et al. 2010	2002–2005 2007–2008	2004–2005 (1) 2007–2008 (3)	PFOA PFOA	(1) (2)	(0/1) (1/1)	10 ng/L 16 ng/L	Assumed steady state. Public water population used for training, bottled water population for testing; all data after intervention in June 2007.
Paulsboro, New Jersey	Graber et al. 2019; Post et al. 2013	2009–2013	2016 (1)	PFOA, PFOS, PFNA, PFHxS	(1)	(1/0)	5 ng/L	Assumed steady state until intervention in April 2014, 2.2 y prior to serum data collection.
Warminster, Pennsylvania	U.S. EPA 2017; Penn DOH 2019; WMA 2018	2013–2014	2018 (1)	PFOA, PFOS, PFNA, PFHxS	(1)	0/1	NR	Assumed steady state until intervention in July 2016, 2 y prior to serum data collection.
Warrington, Pennsylvania	EWG 2019b; Warrington Township 2014, 2018; Penn DOH 2019	2014–2015	2018 (1)	PFOA, PFOS, PFHxS	(1)	0/1	NR	Assumed steady state until intervention in July 2016, 2 y prior to serum data collection.

Note: Table shows location, dates, PFAS, population size, training/testing data set size as individuals or (populations), and limit below which water concentrations were reported as nondetects (LOD). Three identified study populations were excluded (see text). When water concentrations are below detection or reporting limits, a uniform prior distribution between 0 and the concentration limit is used. Additional information regarding water concentrations is in Table S2. LOD, level of detection; NR, not reported; PFAS, poly- and perfluoroalkyl substances; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane sulfonic acid; PFNA, perfluorononanoic acid; PFHxS, perfluorohexane sulfonate.

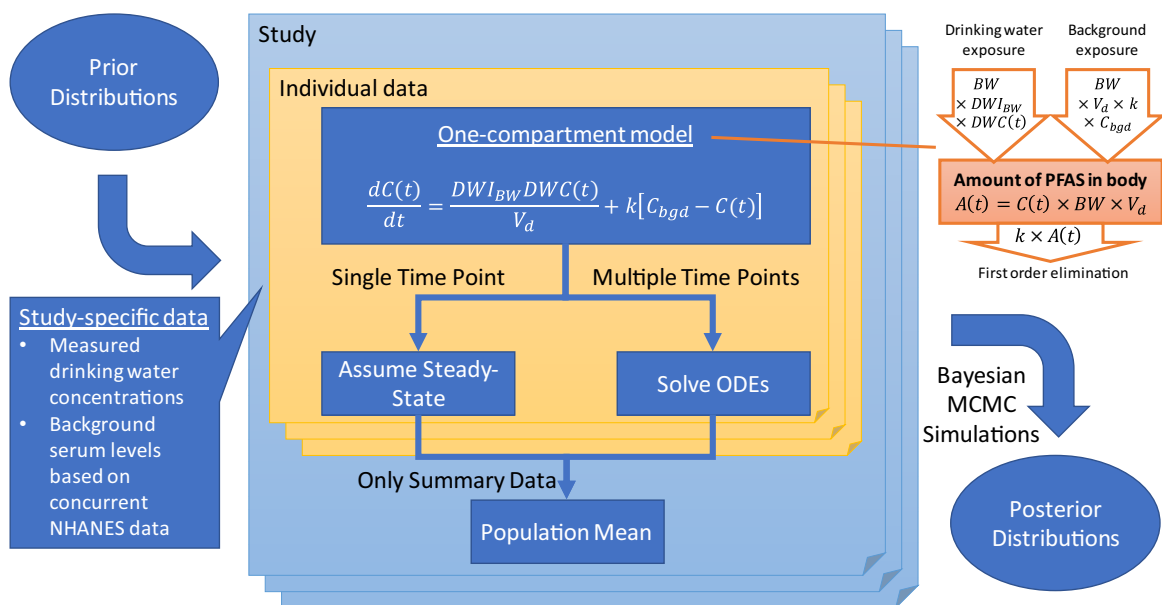


Figure 1. Schematic of modeling approach. A one-compartment pharmacokinetic model in the form of an ordinary differential equation (ODE) for serum concentration $C(t)$ is the basis at the individual level. This model accounts for both drinking water exposure, as well as background exposure, and has parameters of body weight (BW), drinking water intake per unit BW (DWI_{BW}), drinking water concentration (DWC , which can be function of time t), background serum concentration (C_{bgd}), elimination rate k , and volume of distribution V_d . If data are only at a single time point, then steady state is assumed, whereas if there are data at multiple time points, the ODEs are solved numerically. If only summary data are available, then the population mean is predicted for comparison. The Bayesian calibration uses prior distributions for each parameter, some of which are study specific. Markov chain Monte Carlo (MCMC) simulations are used to generate posterior distributions for the population mean and population geometric standard deviation for each parameter. See the “Methods” section for additional details. Note: NHANES, National Health and Nutrition Examination Survey; PFAS, poly- and perfluoroalkyl substances.

For individual data, there are two common situations. In the first, time-course data are available for the individual. In this case, the model sets $t = 0$ at the initial sampling time point, where $C(0) = C_0$. We then evaluate subsequent time points at times after the initial sample point using Equation 1 for time-varying water concentrations and Equation 2 for constant water concentrations. This approach avoids the uncertainty regarding what the background and water concentrations were before the first sampling point but requires data on the water concentrations between the first and last serum sampling point. In the second situation, serum data are available only at a single time point, but historical water concentration data are available. In this case, we use Equation 5, where steady state is assumed until the point of intervention (if any), after which the water concentration level is assumed to be negligible. This simplification allowed us to include data from study locations with no postintervention water concentration data.

If only summary data are available, then the only statistic that the model can directly predict at the population level is the population arithmetic mean because of the presence of the background term. For instance, at steady state, taking the mean of Equation 4, we have the following:

$$\langle C_{bgd+ss} \rangle = \langle C_{bgd} \rangle + \langle C_{ss} \rangle. \quad (8)$$

The C_{bgd} term has a population mean as follows:

$$\langle C_{bgd} \rangle = \exp \left[\mu_{C_{bgd}} + \frac{1}{2} \Sigma_{C_{bgd}}^2 \right]. \quad (9)$$

Because each of the parameters— DWI_{BW} , DWC , k , and V_d —that make up C_{ss} are assumed to be lognormally distributed in the population, their product (and quotient) is also lognormally distributed. Thus, the population mean of C_{ss} is given as follows:

$$\langle C_{ss} \rangle = \left\langle \frac{DWI_{BW} DWC}{k V_d} \right\rangle = \exp \left[\mu_{C_{ss}} + \frac{1}{2} \Sigma_{C_{ss}}^2 \right], \quad (10)$$

where the population GM (e^μ) and geometric standard deviation (GSD) (e^Σ) of C_{ss} are related to those of the other parameters as follows:

$$\mu_{C_{ss}} = \mu_{DWI_{BW}} + \mu_{CDW} - \mu_k - \mu_{V_d} \quad (11)$$

$$\text{and } \Sigma_{C_{ss}}^2 = \Sigma_{DWI_{BW}}^2 + \Sigma_{CDW}^2 + \Sigma_k^2 + \Sigma_{V_d}^2. \quad (12)$$

In the case of an intervention at time Δt prior to the serum measurement, taking the population mean of Equation 5 gives the following:

$$\langle C_{bgd+ss+\Delta t} \rangle = \langle C_{bgd} \rangle + \langle C_{ss} e^{-k\Delta t} \rangle. \quad (13)$$

Because the term $e^{-k\Delta t}$ cannot be expressed exactly in a formula, we applied an approximation. When appearing in the exponential, k is assumed to be normally, rather than lognormally, distributed, with the mean and variance matched to the actual distribution of k . This implies that $e^{-k\Delta t}$ is lognormally distributed, with GM $e^{-(k)\Delta t}$ and GSD $e^{\Delta t \sqrt{\text{VAR}(k)}}$. Then, we make an additional approximation of independence so that each term above consists of a product of independent lognormal distributions. Thus, the product $Y = C_{ss} e^{kt}$ has a population mean as follows:

$$\langle Y \rangle = \langle C_{ss} e^{-k\Delta t} \rangle = \exp \left[\mu_Y + \frac{1}{2} \Sigma_Y^2 \right], \quad (14)$$

$$\text{where } \mu_Y = \mu_{C_{ss}} - \Delta t \times \exp \left(\mu_k + \frac{\Sigma_k^2}{2} \right) \quad (15)$$

$$\text{and } \Sigma_Y^2 = \Sigma_{C_{ss}}^2 + (\Delta t)^2 \times [\exp(\Sigma_k^2) - 1] \exp[2\mu_k + \Sigma_k^2]. \quad (16)$$

A similar approach can be used for summary data consisting of multiple time points as long as the DWC is constant. Taking the arithmetic mean of Equation 2 and separating the terms gives the following:

$$\langle C(t) \rangle = \langle C_{bgd} \rangle + \langle C_0 e^{-kt} \rangle - \langle C_{bgd} e^{-kt} \rangle + \langle C_{ss} \rangle - \langle C_{ss} e^{-kt} \rangle. \quad (17)$$

For the terms $\langle C_0 e^{-kt} \rangle$ and $\langle C_{bgd} e^{-kt} \rangle$, we can use the same approximation from Equations 14–16, replacing the $\mu_{C_{ss}}$ and $\Sigma_{C_{ss}}^2$ with the corresponding values for C_0 and C_{bgd} . We checked the approximations in Equations 14–17 with simulations and found them to have errors mostly of <10% for Δt up to about twice the $T_{1/2}$.

Prior Distributions

Prior distributions for each model parameter are summarized in Table 2 (for additional detail, see Table S3). For the background serum concentrations, C_{bgd} , NHANES data collected closest to the year(s) during which the study data were collected were used as the prior central estimate (Table S3). However, 80% of the appropriate NHANES level was used as C_{bgd} , corresponding to a relative source contribution from drinking water of 20%, so as to avoid double-counting background exposures with exposures due to the measured DWCs. This assumption is consistent with U.S. EPA drinking water guidance, which as a default assumes that 20% of exposure to a contaminant is from drinking water in the absence of chemical-specific data (U.S. EPA 2018). For time-course data, the prior central estimate for the initial concentration C_0 was set to the reported initial value for each individual. For drinking water intake, the population distribution was fixed based on community water source intake data from the Food Commodity Intake Database “What We Eat in America” (FoodRisk 2020) for the years 2005–2010 for those 16–81 years of age (consumers only). For residual error, each data type (individual time-course, individual steady-state, and summary steady-state) involves different assumptions, so each is assumed to have a different residual error.

For elimination rates, we assigned informative prior distributions for each PFAS. We reviewed the available body of literature on $T_{1/2}$ for the four PFAS of interest. We focused on those studies that estimated $T_{1/2}$ from consuming contaminated drinking water. We focused exclusively on nonoccupational populations exposed through contaminated drinking water given that this is a common exposure pathway of relevance to public health and eliminates any potential differences in toxicokinetics associated with high occupational exposure levels. Therefore, we excluded occupationally exposed cohorts without drinking water measurements because ongoing exposures in the community can confound $T_{1/2}$ estimates (e.g., Costa et al. 2009; Olsen et al. 2007). We also excluded studies that estimated elimination from urine alone because total elimination may include other pathways for some PFAS (e.g., Harada et al. 2005; Zhang et al. 2013) and sufficient data were available for estimation without including those studies. However, the results of these studies were considered in setting the bounds on the prior distributions for $T_{1/2}$.

For PFOA, prior distributions were based on the estimates from Bartell et al. (2010) and Seals et al. (2011), the smallest and largest $T_{1/2}$ values identified, respectively. The population mean was centered on the mean log elimination rate [corresponding to a median $T_{1/2}$ of 4.6 y with uncertainty GSD = 1.5, so the 95% confidence interval (CI) of 2.1 to 10.2 covers both studies]. Population variation was based on the reported 95% interval of individual $T_{1/2}$ from Bartell et al. (2010) (log SD = 0.286). Assuming the precision Σ^{-2} has a prior uncertainty coefficient of variation (CV) = $\alpha^{-1/2}$ = 33%, this implies a shape parameter α = 9. The rate parameter β is then derived by matching the prior mean = α/β = 0.286⁻² ≈ 12. For PFOS, this is based on the range of several studies, centered on a GM of 4.8 from Olsen et al. (2007) and the same GSD as PFOA. Based on this range, we dropped a study that found a $T_{1/2}$ of 27 y for PFOS in adults >50 years of age (Zhang et al. 2013) given that it is an outlier compared with the other studies. For population variation, because PFOA has the most data, we used the posterior from PFOA as the prior for PFOS and the other PFAS. For PFNA and PFHxS, the prior central estimates for the $T_{1/2}$ were 4.3 y (from Zhang et al. 2013) and 5.3 y (from Li et al. 2018), respectively. Limited $T_{1/2}$ data were available for PFNA; therefore, we used the

Table 2. Model parameters and prior distributions used in Bayesian parameter calibration.

Parameter (units)	Description	Fixed value or prior for population GM = e^μ	Fixed value or prior for population GSD = e^Σ	Comments (see text for details)
C_{bgd} (μg/L)	Background serum concentration	$\mu \sim \text{LN}(GM_i, 1.5)$	1.2	Year-specific prior (see Table S3).
C_0 (μg/L)	If >1 time point, initial serum concentration	$\mu \sim \text{LN}(GM_{indiv}, 1.5)$	1.2	Individual-specific prior.
DWI_{BW} (mL/kg per day)	Drinking water intake (body weight-adjusted)	12.33	2.43	Fixed, ages 16–81 y, consumers only.
k (per year)	Elimination rate constant	$\mu \sim \text{LN}(GM_{PFAS}, 1.5)$	PFOA: $\Sigma^{-2} \sim \Gamma(9, 0.75)$ Other PFAS: based on PFOA posterior	PFAS-specific prior (see text).
V_d (L/kg)	Volume of distribution	$\mu \sim \text{LN}(GM_{PFAS}, 1.3)$	$\Sigma \sim \text{HN}(0, 0.2)$	PFAS-specific prior (see text).
$GSD_{err,t}$, $GSD_{err,ss}$, $GSD_{err,sum}$	Residual error for individual time-course data (t) or steady-state data (ss), and summary steady-state data (sum)	$GSD_{err} \sim \text{Lunif}(1.1, 10)$	NA	Different data types have different assumptions, so are assumed to have different residual errors.
$DWC_{<MRL}$ (μg/L)	Drinking water concentration when below minimum reporting level (MRL)	$DWC_{<MRL} \sim \text{Unif}(0, \text{MRL})$	NA	If DWC is below reporting level, assume uniform distribution from 0 to reporting level.

Note: The lognormal distribution is specified by LN (GM, GSD). The gamma distribution is specified by $\Gamma(\alpha, \beta)$ for shape and rate parameters α and β , respectively. The half-normal distribution is specified by HN (M, SD) and is defined only by positive values. The log-uniform distribution is specified by Lunif (min, max), and the uniform distribution is specified by Unif (min, max). MRL is the minimum reporting level in the drinking water testing. DWC, drinking water concentration; Err, error; GM, geometric mean; GSD, geometric standard deviation; HN, half-normal distribution; indiv, individual; LN, lognormal distribution; Lunif, log-uniform distribution; max, maximum; min, minimum; NA, not applicable; PFAS, poly and perfluoroalkyl substances; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane sulfonic acid; Unif, uniform distribution.

data from Zhang et al. (2013) to center our prior distribution even though these data were based on blood–urine pairs. We chose to use the higher estimate (4.3 in all males and females >50 years of age) vs. the lower (2.5 y in younger females) as our central estimate for the prior distribution given that the elimination rate is hypothesized to be longer for the longer-chained PFAS (PFNA is a 9-carbon chain PFAS) (Graber et al. 2019). For PFHxS, we chose to center our estimate on the study by Li et al. (2018) that the MDH used in their PBPK model for PFHxS. $T_{1/2}$ for PFHxS also ranged widely in the literature, from 4.7 y (females only) to 15.5 y (Li et al. 2018; Worley et al. 2017a), when excluding the urine-based outlier of 35 y from Zhang et al. (2013).

We based V_d priors on the same literature search as was conducted for $T_{1/2}$. The prior distribution for the population GM was centered on 0.17 L/kg for PFOA and 0.23 L/kg for PFOS, based on Thompson et al. (2010) with uncertainty GSD = 1.3 for both PFOA and PFOS. The prior distribution for population variation was based on a weakly informative half-normal prior (per a recommendation by Gelman 2006) with mean of 0.16. For PFHxS, the prior central estimate was 0.25 L/kg, based on Sundström et al. (2012), which found a V_d range of 0.2 to 0.3 L/kg informed by rat, mice, and monkey data, and Koponen et al. (2018), which assumed the same V_d for PFHxS as for PFOS. Owing to a lack of data, the prior central estimate for PFNA was assumed to be the same as for PFOA (0.17 L/kg), consistent with the assumption made in Koponen et al. (2018), which was informed by rodent data.

For some studies, the DWCs are reported to be “below the minimum reporting level.” In this case, a uniform distribution uncertainty between zero and the reporting level is used as a prior for the actual DWC.

Model Implementation and Evaluation

Data were separated into “training” and “testing” data sets, with only training data used for calibration (i.e., part of the likelihood function). For studies with individual data, training data consisted of half the individuals of each sex, randomly selected, with the remaining individuals treated as testing data sets. For studies with summary data only, half of the studies were selected for training and half for testing. In both cases, if an odd number of individuals or studies were present, the additional case was randomly assigned to either testing or training.

We implemented the model in the open-source software MCSim (version 6.1.0; GNU MCSim; <https://www.gnu.org/software/mcsim/>), and all analyses were performed in R within RStudio (R Development Core Team; version 3.6.1; Rstudio Team). Plotting and summarization were performed with tidyverse packages (Wickham et al. 2019). Four independent Markov chain Monte Carlo (MCMC) chains were run for each PFAS. We assessed convergence using the potential scale reduction factor ($\hat{R}$), which approaches 1.0 with convergence, and for which a value of ≤ 1.2 is proposed as acceptable (Gelman et al. 2013). As mentioned above, PFOA was run first, because it has the most individual data. The posterior distribution for the population variance of the elimination rate for PFOA was used as the prior distribution for this parameter for the other PFAS (i.e., replacing the inverse gamma distribution prior used for PFOA). We fit the PFOA posterior for this variance parameter to a log-normal distribution. R codes for all modeling and analyses are included in supplementary materials and available at <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Sensitivity Analysis

We performed multiple local sensitivity analyses with respect to our modeling assumptions and approach to validate the parameter

estimates produced by the model. These included *a*) changing the relative source contribution from drinking water in background serum concentration from 20% to either 0% or 80%, *b*) shifting the prior distributions for V_d to 20% above or below the primary estimate, and *c*) evaluating the effect of using alternative training vs. testing data sets. Specifically, we analyzed the effect of using only individual-level serum concentrations, using only population-level aggregates, and switching the training and testing data sets from the primary analysis. Finally, we performed an analysis focused only on parameter estimation, where all data were used for training. In all cases, we reran the entire model calibration and evaluation process using the alternative parameter values, prior distributions, or data sets.

Results

Numeric outputs and data that are not under a data sharing agreement are available in the Supplemental Information. Individual-level data for Decatur, Alabama, cannot be published in detailed form.

Serum and Water Data

Comprehensive details on the identified populations and studies are provided in Table 1 and in Table S1. Briefly, for the Decatur population ($n=37$ after excluding eight individuals owing to occupational history), the Agency for Toxic Substances and Disease Registry (ATSDR) supplied the individual serum data. For the Arnsberg population, the study by Hölzer (2008) provided individual serum PFOA levels graphically for 2006 and 2007 ($n=151$). We derived and estimated those values from the line graph provided in the study to use them as individual serum data inputs in the model at two time periods. For the Minnesota population, the study by Johnson et al. (2017) provided individual-level serum data for PFOS and PFOA correlated with individual drinking water exposure ($n=98$). These were also derived from digitizing figures with WebPlotDigitizer (version 4.2) because tabular data were not provided. We obtained all other data from text or tables from study publications. Individual water consumption information was not available in any of the identified studies.

A number of identified populations were excluded because the publications were missing critical information. We excluded the North Wales, Pennsylvania, cohort because the water data from before the serum measurements were not available and because some of the water was purchased from a neighboring town (North Wales Water Authority 2018). We excluded the Ronneby cohort because serum concentrations were not reported at individual time points (Li et al. 2018). The Uppsala County cohort was excluded because it contained only a cumulative estimate of months of exposure for children. For the Arnsberg cohort, we included only data for adults, given that it was the only study that separated children’s serum levels from adults’.

The study populations we used for modeling, along with summary statistics of the serum and water concentrations for each PFAS, are summarized in Table 1. In some cases, there was only a single water or serum level for an individual or a population. In other cases, there were multiple water levels preceding the serum level, or ideally, more than one serum level with corresponding water concentrations over time at the individual level.

Model Convergence and Fit

Parameter estimates in all four models achieved excellent convergence ($\hat{R} < 1.05$) with a reasonable number of MCMC iterations (4 chains; 20,000 iterations per chain). The resulting overall model fits comparing posterior median predictions and data are shown in Figure 2. For training data (Figure 2, left panels), the

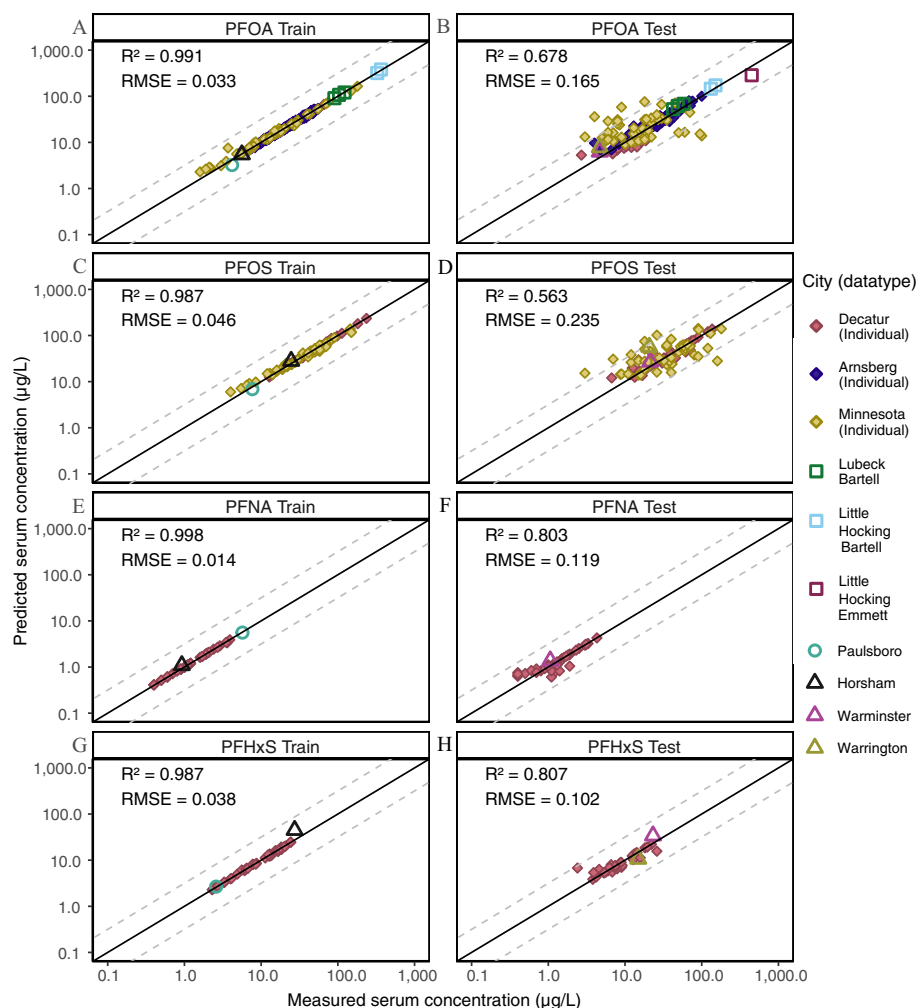


Figure 2. Overall evaluation of model fit. Comparison of data and median posterior predictions for (A,B) perfluorooctanoic acid (PFOA), (C,D) perfluorooctane sulfonic acid (PFOS), (E,F) perfluorononanoic acid (PFNA), and (G,H) perfluorohexane sulfonate (PFHxS) for both (A,C,E,G) training data and (B,D,F,H) testing data. The solid line represents equality, and the dashed line represents a 3-fold error. In each panel, R^2 and root mean square error (RMSE) are also shown in log₁₀ units. The underlying numeric values can be found in Table S5.

model was able to match the data very tightly. The GSD_{err} for individual time-course data was ~ 1.1 for all four PFAS, indicating that the model fit had a residual error (difference between predictions and training data) of only $\sim 10\%$ when time-course data were available. In cases where data are only available at a single time point per individual, such as for the Minnesota data for PFOA and PFOS, the GSD_{err} was ~ 1.5 , indicating $\sim 50\%$ residual error. A larger error such as this one is expected because of the need to approximate steady state in these cases. Moreover, the Minnesota data appeared to tend toward overprediction, which is consistent with the use of a steady-state approximation (i.e., in reality, steady state would not have been reached, so the actual concentration would tend to be lower). Summary time-course data, available only for PFOA, had a GSD_{err} of ~ 1.2 , indicating $\sim 20\%$ error. For summary data at a single time point, however, the residual errors were quite a bit larger: ~ 2 -fold for PFOA and PFOS, 2.3-fold for PFNA, and 2.8-fold for PFHxS. Remarkably, however, the residual error in all cases was < 3 -fold. These results show the importance of individual data for accurately estimating PFAS elimination.

The comparisons with the testing data (Figure 2, right panels) showed a similar trend. For individual data, the model performed best when we used data sets containing individual time-course data, with residual errors well within 50%. Individual data with a

single time point and summary steady-state data performed worse, although residual errors were generally within 3-fold. Figure 3 shows the posterior distributions of predictions for the Decatur time-course data, which were available for all four PFAS. For both training and testing data sets, the data were well within the CIs of the posterior predictions. Results were similarly accurate for the individual data sets across all PFAS (Figures S1–S4). However, for summary data, the predictions were less accurate for both the training and testing data sets.

Posterior Distributions

Posterior distributions for the main toxicokinetic parameters of $T_{1/2}$ and V_d , as well as the derived parameter of clearance, are shown in Table 3. These reflect updating of the prior distributions after consideration of the likelihood of the data. In addition to estimates for the population GM and population GSD, the model makes a prediction for a “random individual,” which combines uncertainty and variability. The random individual is relevant to the general public because for any individual, there are two sources of uncertainty: *a*) the uncertainty in the population distribution (GM and SD), and *b*) uncertainty in where one is located on the population variability distribution (e.g., does an individual have a higher or lower than typical $T_{1/2}$ for a PFAS?).

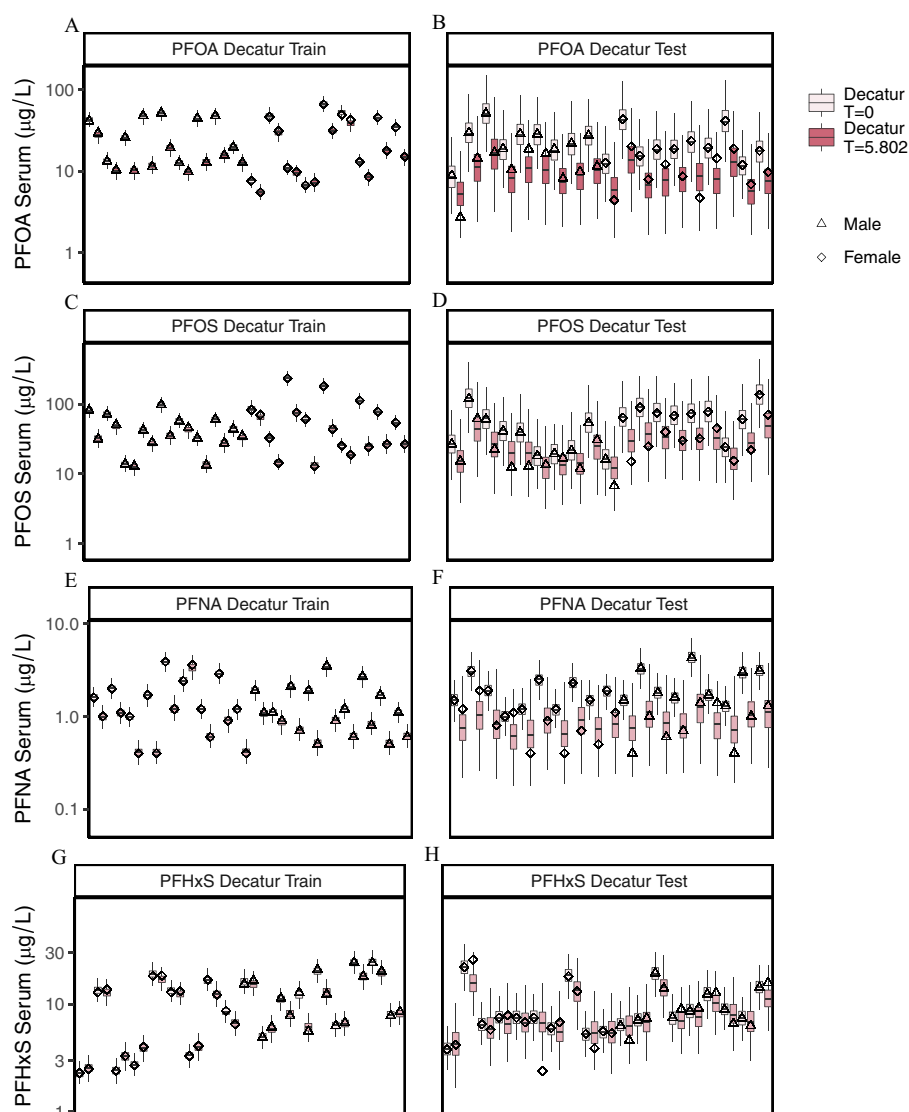


Figure 3. Data and posterior distribution of predictions for Decatur, Alabama, data. Comparison and Decatur data (symbols) and distribution of posterior predictions (box plots) for (A,B) perfluorooctanoic acid (PFOA), (C,D) perfluorooctane sulfonic acid (PFOS), (E,F) perfluorononanoic acid (PFNA), and (G,H) perfluorohexane sulfonate (PFHxS) for both (A,C,E,G) training data and (B,D,F,H) testing data. Shading indicates time (T), in years, since the first serum sample. Samples from the same individuals are paired such that the two left-most bars indicate samples from one individual taken at two time points (the initial time point is $T = 0$, and the second is $T = 5.802$ y later). The underlying numeric values can be found in Table S6.

Comparisons of prior and posterior distributions for the $T_{1/2}$ are shown in Figure 4. For all four PFAS, the posteriors for the population GMs were noticeably shifted and narrower than the prior distributions. This indicates that the data were informative relative to the prior. For PFOA, the posterior for the population GSD was also shifted and indicated more population variation in the $T_{1/2}$ than under prior assumptions. However, for the other PFAS, the priors and posteriors for the population variation were similar because there were insufficient numbers of individuals to inform this parameter; in other words, the data were not informative relative to the prior. Nonetheless, this indicates that the data are consistent with the amount of population variation in $T_{1/2}$ observed with PFOA. $T_{1/2}$ estimates (95% CIs) for the population GM were 3.14 (2.69, 3.73) y for PFOA, 3.36 (2.52, 4.42) y for PFOS, 2.35 (1.65, 3.16) y for PFNA, and 8.30 (5.38, 13.5) y for PFHxS. For PFHxS, there was a noticeable shift from the prior to the posterior, with the central estimate moving from 5.29 to 8.30 y. Although the uncertainty ranges in Figure 4 look similar on the

natural scale, the 95% CI range on the log-scale shrank substantially, from 4.9- to 2.5-fold.

Comparisons of prior and posterior distributions for the V_d are shown in Figure 5. V_d posterior estimates ranged from 0.19 to 0.428 L/kg across the four PFAS. In general, the posterior estimates for the V_d across the four PFAS are shifted to larger values as compared with the prior distributions, indicating that the data were informative for the population mean of V_d . A larger V_d indicates that for a given intake, the ratio between DWCs and serum concentrations would tend to be larger as compared with previous studies; this result is likely due to accounting for background (nondrinking water sources). There are insufficient data, however, to substantially inform the population variability in the V_d given that the prior and posterior distributions are similar.

With respect to clearance, which is the product of the rate coefficient k and the V_d , posterior medians ranged from 0.025 to 0.095 L/kg per year, with PFHxS < PFNA < PFOS < PFOA. These values tended to be on the higher end of the prior

Table 3. Summary of posterior distributions identified through Bayesian parameter calibration.

Parameter [prior population GM median (95% CI)]	Population GM median (95% CI)	Population GSD median (95% CI)	Random individual median (95% CI) [98% CI] ^a
PFOA			
Half-life (y) [4.6 (2.1, 10.2)]	3.14 (2.69, 3.73)	1.57 (1.42, 1.73)	3.13 (1.13, 7.83) [0.90, 9.14]
Volume of distribution (L/kg) [0.17 (0.10, 0.28)]	0.43 (0.32, 0.59)	1.12 (1.01, 1.47)	0.43 (0.27, 0.74) [0.23, 0.87]
Clearance (L/kg per year) [0.037 (0.014, 0.095)]	0.095 (0.074, 0.126)	1.62 (1.45, 1.85)	0.097 (0.0369, 0.262) [0.0327, 0.341]
PFOS			
Half-life (y) [4.8 (2.2, 10.6)]	3.36 (2.52, 4.42)	1.57 (1.42, 1.76)	3.40 (1.28, 8.42) [1.20, 9.96]
Volume of distribution (L/kg) [0.23 (0.14, 0.38)]	0.32 (0.22, 0.47)	1.10 (1.01, 1.38)	0.32 (0.19, 0.51) [0.15, 0.56]
Clearance (L/kg per year) [0.048 (0.019, 0.123)]	0.066 (0.048, 0.092)	1.60 (1.45, 1.83)	0.066 (0.0245, 0.176) [0.0203, 0.199]
PFNA			
Half-life (y) [4.3 (1.9, 9.5)]	2.35 (1.65, 3.16)	1.53 (1.40, 1.70)	2.27 (0.83, 5.36) [0.76, 5.94]
Volume of distribution (L/kg) [0.17 (0.10, 0.28)]	0.19 (0.11, 0.30)	1.12 (1.01, 1.51)	0.18 (0.10, 0.32) [0.09, 0.40]
Clearance (L/kg per year) [0.040 (0.015, 0.102)]	0.056 (0.033, 0.093)	1.57 (1.42, 1.86)	0.056 (0.019, 0.163) [0.0165, 0.199]
PFHxS			
Half-life (y) [5.3 (2.4, 11.7)]	8.30 (5.38, 13.5)	1.57 (1.42, 1.77)	8.12 (2.96, 21.6) [2.19, 24.8]
Volume of distribution (L/kg) [0.25 (0.15, 0.42)]	0.29 (0.17, 0.45)	1.11 (1.00, 1.45)	0.28 (0.16, 0.46) [0.14, 0.56]
Clearance (L/kg per year) [0.047 (0.018, 0.122)]	0.025 (0.012, 0.039)	1.61 (1.45, 1.86)	0.022 (0.0075, 0.078) [0.0065, 0.10]

Note: Model parameters are shown by chemical species with GMs, GSDs, and CIs. CI, confidence interval; GM, geometric mean; GSD, geometric standard deviation; PFHxS, perfluorohexane sulfonate; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane acid; PFNA, perfluorononanoic acid.

^a98% CI shows the variation from the 1st percentile random individual to the 99th percentile random individual.

distributions for PFOA, PFOS, and PFNA, and toward the lower end for PFHxS. We also found that posterior distributions for the components of clearance, $T_{1/2}$, and V_d were uncorrelated (all $R^2 < 0.05$). Posteriors for the population variation were slightly larger than those for $T_{1/2}$, reflecting the relatively smaller contribution of V_d to interindividual variation.

To check for the possibility of systematic biases, we examined the posterior distributions for individual k and V_d parameters and compared them across study cohorts. For instance, if individual posteriors for one location were systematically different from individual posteriors for another location, then that would suggest errors in the model or parameters. As shown in Figure S5, there is no discernable difference across cohorts from different cities/locations, and all individual parameter posterior samples had z -scores that were statistically consistent with the expected standard normal distribution.

Sensitivity Analysis

The posterior estimates for parameters were generally stable across all sensitivity analyses, and in all cases, there was substantial overlap between the 95% CI from each sensitivity analysis and that from the primary analysis (Table S4, Figures S6–S9). Changing relative source contribution from drinking water and nondrinking water sources had little effect on $T_{1/2}$ or V_d estimates. Swapping the data sets used for testing with those used for training produced nearly identical results to the primary analysis. Increasing and decreasing the priors for V_d resulted in nearly identical posterior estimates for all parameters. The largest differences from the primary analysis occurred when removing either the individual- or the population-level data for training. The posteriors for the GSD for variability in $T_{1/2}$ were much lower when we used only population summary data, as would be expected

because it is more challenging to estimate variability when only summary data are available.

Discussion

To our knowledge, ours is the largest analysis to date of individual serum data of communities with known and measured PFAS drinking water contamination. We incorporated data from 13 studies performed across widespread geographic locations. We have integrated these multiple data sets in a Bayesian toxicokinetic analysis to estimate the $T_{1/2}$ and V_d , as well as the population variability, for four common PFAS. We have also incorporated NHANES estimates of background exposures over time, without which kinetic parameter estimates may be biased. Our model accurately predicts serum data from a large number of individuals across multiple studies, including data not used for calibration. Furthermore, the posterior estimates are insensitive to a variety of changes to the prior inputs and to the design of testing and training data sets, suggesting these estimates are stable given the current data.

Our results for the population GM of $T_{1/2}$ are in the range of several previous studies that we did not use in our analysis. For instance, Olsen et al. (2007) estimated $T_{1/2}$ (95% CIs) of occupationally exposed retirees to be 3.8 (3.1, 4.4) y for PFOA, 5.4 (3.9, 6.9) y for PFOS, and 8.5 (6.4, 10.6) y for PFHxS. The values for PFOA and PFOS are somewhat longer than our estimates, but Olsen et al. (2007) did not account for continued exposure due to drinking water contamination, which was later found to be substantial in the community. Li et al. (2018) reported results from an analysis of residents in Sweden exposed via drinking water contamination and reported shorter $T_{1/2}$ (95% CIs) of 2.7 (2.5, 2.9) y for PFOA, 3.4 (3.1, 3.7) y for PFOS, and 5.3 (4.6, 6.0) y for PFHxS, which are concordant with our estimates for PFOA and

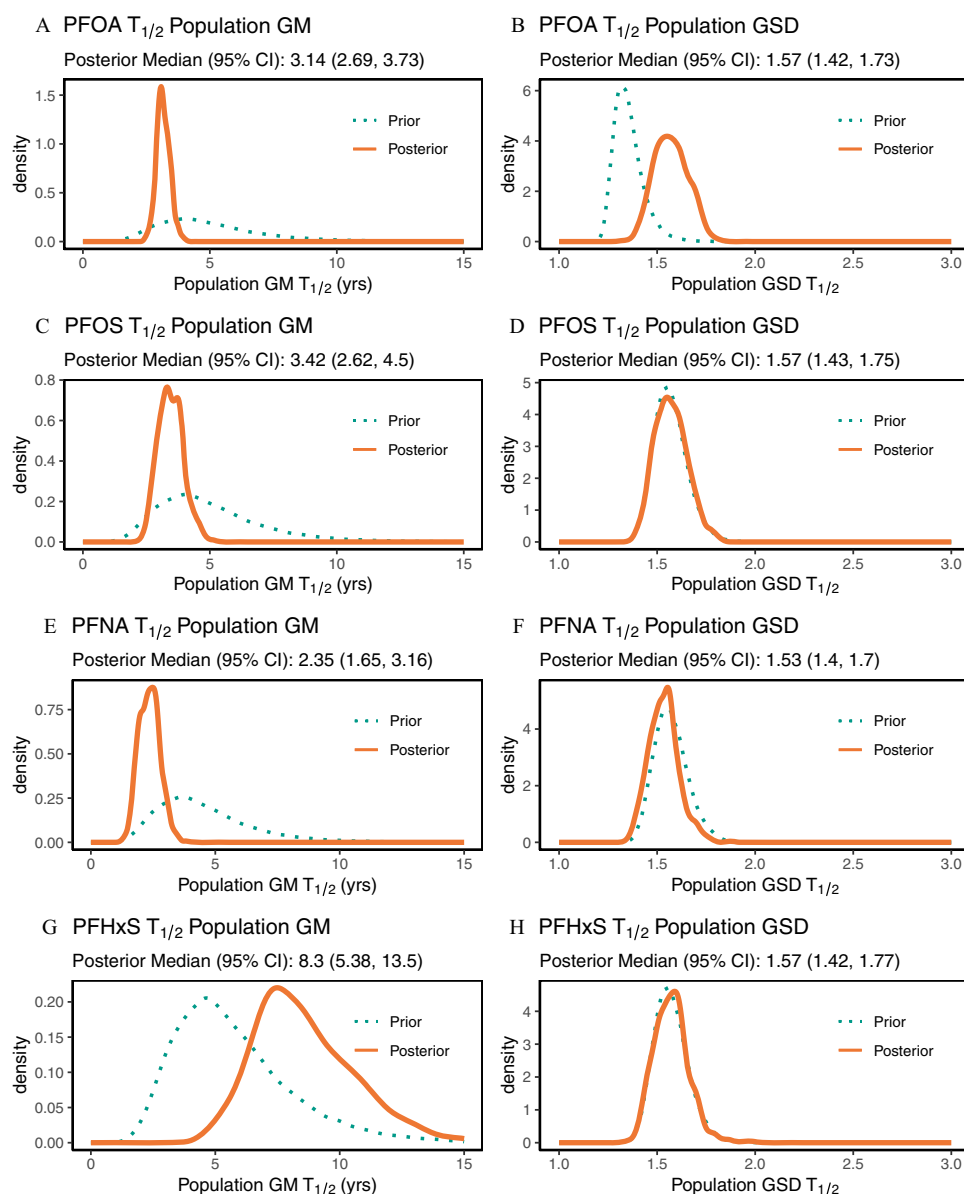


Figure 4. Prior and posterior distributions for half-life ($T_{1/2}$). Comparison of priors (cyan dotted lines) and posteriors (orange lines) for (A,C,E,G) $T_{1/2}$ population geometric mean (GM) and (B,D,F,H) population geometric standard deviation (GSD) for (A,B) perfluorooctanoic acid (PFOA), (C,D) perfluorooctane sulfonic acid (PFOS), (E,F) perfluorononanoic acid (PFNA), and (G,H) perfluorohexane sulfonate (PFHxS). Text includes posterior median and confidence intervals (CI). The underlying numeric values are presented in [Tables 2 and 3](#) and in the text.

PFOS but somewhat shorter than our estimates for PFHxS. Overall, our analysis supports the higher estimate for PFHxS $T_{1/2}$, though recognizing potential for substantial population variation.

Our results for the population GSD are also within the range of population variation reported in previous studies that we did not use in our analysis. Population variation in individual $T_{1/2}$ from Olsen et al. (2007) was estimated to be ~ 1.49 for PFOA, 1.66 for PFOS, and 1.78 for PFHxS. The values for PFOA and PFOS are similar to those found in our analysis, with the value for PFHxS somewhat higher. Variation reported by Li et al. (2018) was somewhat less, with a GSD of ~ 1.4 across all PFAS, at the low end of the CI from our analysis.

Our estimates for the V_d of PFOA and PFOS are somewhat larger than values reported in the literature and used as priors in the analysis (Figure 5). For instance, Thompson et al. (2010) found 0.17 L/kg for PFOA; we found 0.43 L/kg (95% CI: 0.32, 0.59). Thompson et al. (2010) found 0.23 L/kg for PFOS, and our prior

estimate was slightly larger at 0.32 L/kg (95% CI: 0.22, 0.47). Our findings for PFNA [0.19 L/kg (95% CI: 0.11, 0.30)] and PFHxS [0.29 L/kg (95% CI: 0.17, 0.45)] were similar to priors (Figure 5). Importantly, none of the previous values used to develop prior estimates for V_d (see the “Methods” section) were based on statistical calibration using multiple data sources. Previous values were estimated based on adjusted values from animal studies (Sundström et al. 2012) or other PFAS (Koponen et al. 2018) or were based on assumed serum and water data from some of the same cohorts as used here (e.g., Lubeck, West Virginia; Little Hocking, Ohio) but with fixed values for other parameters, such as drinking water rates (Thompson et al. 2010). In addition, we made a greater effort to adjust for background exposures compared with previous studies; underestimating background exposure can lead to underprediction of V_d . For a given $T_{1/2}$, DWC, and observed serum concentration, underestimating background will require a smaller V_d to fit the observation. Overall, our Bayesian approach provides a

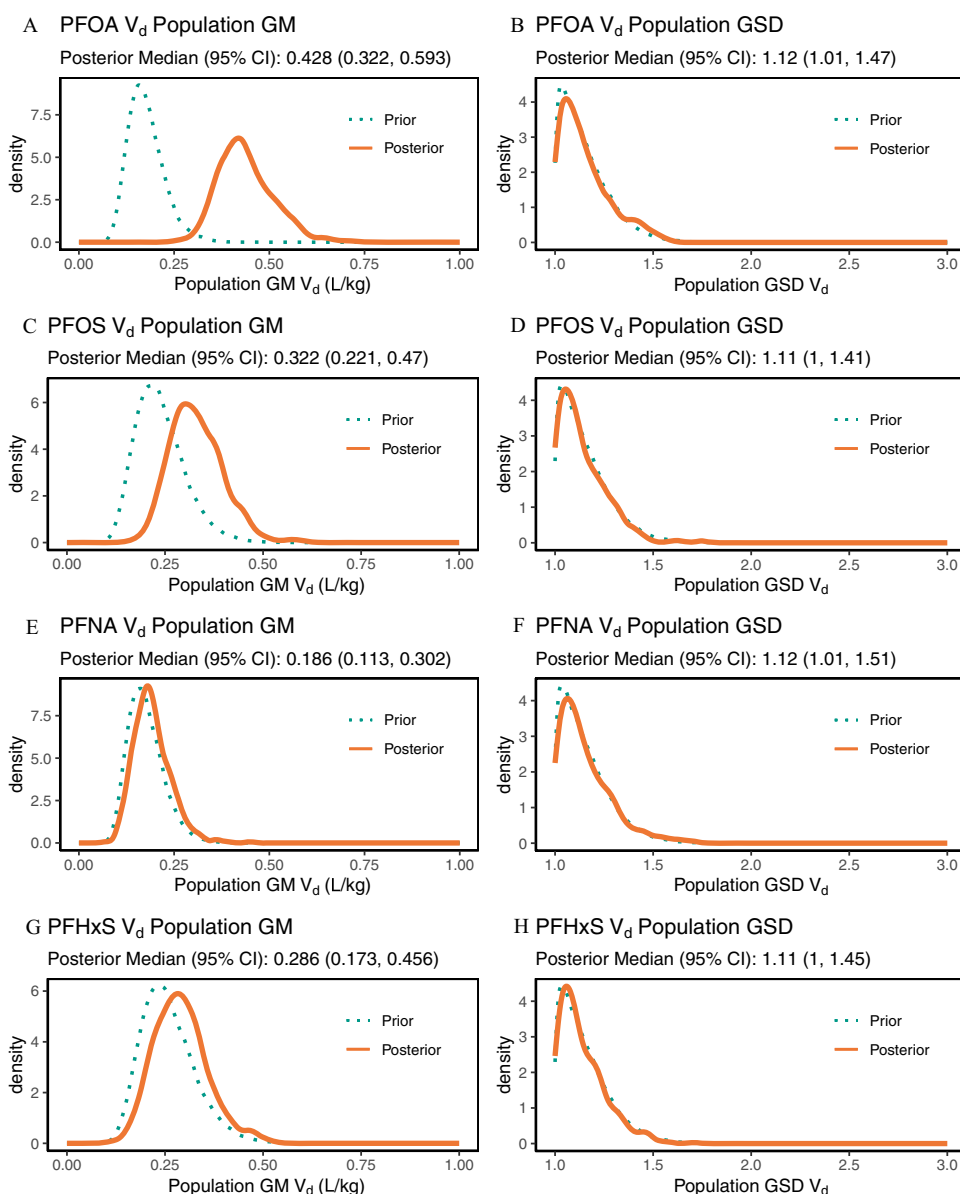


Figure 5. Prior and posterior distributions for the volume of distribution. Comparison of priors based on literature (cyan dotted lines) and posteriors (orange lines) for the volume of distribution (A,C,E,G) population geometric mean (GM) and (B,D,F,H) population geometric standard deviation (GSD) for (A,B) perfluorooctanoic acid (PFOA), (C,D) perfluorooctane sulfonic acid (PFOS), (E,F) perfluorononanoic acid (PFNA), and (G,H) perfluorohexane sulfonate (PFHxS). Text includes posterior median and confidence intervals (CIs). The underlying numeric values are presented in [Tables 2 and 3](#) and in the text.

rigorous basis for V_d estimates because it integrates multiple individual-level data sets, includes extensive prior information on background exposures, and incorporates population variability.

In addition, we have derived posterior estimates for both the population GM and population variation in clearance, which is the key parameter for chemical-specific values for interspecies extrapolation and interindividual variability, as well as for *in vitro* to *in vivo* extrapolation (IVIVE). For instance, as discussed in WHO IPCS (2005) and U.S. EPA (2014) guidance documents, the ratio of clearances can be used to replace default uncertainty factors for interspecies and interindividual toxicokinetic differences. For interspecies extrapolation, these results could be combined with those of Wambaugh et al. (2013) for extrapolating experimental animal points of departure to human equivalent doses. In addition, for interindividual variability, the ratio of the median to 1% random individual from [Table 3](#) could replace the default value of $UF_{H,TK} = 3.16$ for toxicity end points in (nonpregnant) adults.

Interestingly, the ratios of the median to the 1% random individual fall within a narrow range—3.0 for PFOA, 3.3 for PFOS, and 3.4 for PFNA—and near to the default value for PFHxS. Finally, for IVIVE, clearance estimates can be used to convert from *in vitro* test concentrations to oral equivalent doses so as to put high-throughput screening results in the context of human exposure (Wetmore et al. 2015).

One limitation of our analysis is that our estimates are not specific to age, sex, geographic location, or race/ethnicity, nor do we have geography-specific estimates of background exposures. We did not identify sufficient data to adequately stratify our analysis among these different groups, so we cannot assess the potential for such differences to affect our results. In Arnsberg, the women were all mothers, and we did not have information on their ages. In Decatur, Alabama, the mean age in 2010 was 52 y, and in 2016 the mean age was 63 y, with very few younger women. For the Minnesota data set, we did not have age or sex

information for the individuals. Studies have shown that PFAS serum levels can vary by age, sex, race/ethnicity, and geographic location (Park et al. 2019). In addition, Park et al. (2019) found that parity and menstrual bleeding were important predictors of PFAS levels. This is consistent with other studies in both animals and humans, which indicate that serum levels in menstruating women, and in women who have breastfed or given birth, may be lower than in older women and men (Brantsæter et al. 2013; Huang et al. 2019; Singer et al. 2018). Further, Zhang et al. (2013) found shorter $T_{1/2}$ in younger women compared with older women and men, hypothesizing that menstrual clearance is important for PFOS, PFNA and PFHxS and less pronounced for PFOA. Future work as more individual serum-level data become available could better distinguish these differences.

This work has a number of additional limitations. First, it excludes some studies for which individual serum data could not be readily obtained, or for which corresponding water concentration information was not available (Table S1). This limitation is particularly important for PFNA and PFHxS, for which individual data were available only for the Decatur cohort, where PFAS drinking water levels were below the minimum reporting level. This limitation is less important for PFOA and PFOS, for which more than one study with many individuals is included in the analysis. A corollary limitation is that because only three studies had individual data for PFOA, two studies for PFOS, and one study for PFNA and PFHxS, we could not use separate studies for training and testing, but instead split individuals within each data set. Thus, the training and testing data sets were not completely independent. A sensitivity analysis switching testing and training data sets gave similar results (Figures S6–S9), which could indicate either an issue with independence or robust estimates. A further comparison of individual-level V_d and k posteriors for independent study cohorts of PFOA- and PFOS-containing individual-level data showed no systematic differences across cohorts, as expected for physiological parameters (Figure S5). Thus, the nonindependence of testing and training data sets is not a critical limitation for use of these results. Our main objective here was to estimate model parameters.

Second, for many studies, particularly those with summary data, we had to make a steady-state assumption of constant DWCs in the past based on few measurements, or assume negligible concentrations after drinking water interventions due to missing post-intervention measurements. This limitation is again more acute for PFNA and PFHxS but is also an issue for PFOS, for which the Decatur cohort was the only data set with time-course data. Thus, we have greatest overall confidence in the PFOA results, moderate confidence in the PFOS results, and greater uncertainty in the PFNA and PFHxS results. Obtaining additional individual-level time-course data, including postintervention concentrations, would be the most effective way to address these limitations. This analysis also did not consider the potential contribution of exposure to precursors or degradation products of the PFAS.

Finally, although a one-compartment model has the virtue of simplicity and ease of implementation, the results are ultimately empirical. They do not provide mechanistic insights, nor can they be used to characterize tissue-specific internal dose. Furthermore, they cannot directly account for saturable reabsorption mechanisms; however, unlike exposures in experimental animal studies, human drinking water exposure levels appear to be well below saturation, so reabsorption can be lumped into an overall first-order clearance process. In particular, the concentrations in serum in the studies analyzed here were well below any saturation, and hence an explicit model for this mechanism is not necessary. For example, the K_M values for organic anion transporters in the kidney range from 20 to 78 mg/L for PFOA (Worley et al. 2017b;

Nakagawa et al. 2008). This contrasts with values from experimental animal studies, which are performed at higher exposures and have lower degrees of renal reabsorption, where such a mechanism is needed (e.g., Andersen et al. 2006). In any case, given the need to incorporate individual-level data and complex exposure patterns, this model is a useful first step in combining data from many sources while also estimating interindividual variability. Moreover, given the relatively high data needs for even this one-compartment analysis—extensive individual serum data along with local DWCs—it will be challenging to validate more complex toxicokinetic models for PFAS more generally.

These results have a number of important public health implications. The degree of variability in $T_{1/2}$ across individuals means that for the same external exposure, some individuals will experience much greater internal exposure than others and, therefore, have a higher effective dose (Table 3). For instance, the 95% CI of the $T_{1/2}$ for PFOA of a random individual drawn from the population spans from 1.13 to 7.83 y, with the 98% CI spanning a 10-fold range from 0.90 to 9.14 y. Across the four PFAS, the range from the 1st to the 99th percentile of $T_{1/2}$ for a random individual span between 8- and 11-fold (Table 3). In addition, the higher estimate for the V_d as compared with previous studies implies that, for a given serum concentration, the body burden is greater. Although biomonitoring data on selected PFAS suggest that levels are declining across the population overall, drinking water contamination continues to be an issue in many parts of the country. Better estimates of toxicokinetic parameters of PFAS, as well as their variation in the population, will be essential in better characterizing the potential public health effects of PFAS, and the methods we applied here can be readily applied to other PFAS where individual time-course data are available for both serum and water concentrations. Ultimately, these estimates will be essential in supporting risk assessments and risk management decisions aimed at reducing current and future exposures to this ubiquitous and persistent class of compounds.

Acknowledgments

Abt Associates authors (M.T.L., C.L., A.A., D.M., and S.S.) and W.A.C. were supported under contracts PO0400247 and GS00F045DA (both to M.T.L.) from the Agency for Toxic Substances and Disease Registry (ATSDR) under a subcontract to Guidehouse LLC. This work was also supported, in part, by grants from the National Institutes of Health/National Institute of Environmental Health Sciences (P42 ES027704, P30 ES029067, both to W.A.C.). Preparation of this paper was also supported by Abt Associates internal funds. The paper was improved by presentation at an Abt Work in Progress Seminar and comments received as part of that process. P. Do and L. Katz at Abt Associates provided valuable support on the literature review, while D. Ferguson and R. Balachandran provided technical editing. C. Welsh provided helpful review comments as the program lead for the Computational Toxicology and Methods Development Laboratory at the ATSDR. M. Shoemaker, F. Sieling, S. Lane, and L. Pogorelov at Guidehouse LLC, as well as D. Hunt, M. Lorie, and E. Chen of Abt Associates, provided project management support.

The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the Centers for Disease Control and Prevention/the Agency for Toxic Substances and Disease Registry.

References

- Andersen ME, Clewell HJ III, Tan YM, Butenhoff JL, Olsen GW. 2006. Pharmacokinetic modeling of saturable, renal resorption of perfluoroalkylacids in monkeys—probing the determinants of long plasma half-lives. *Toxicology* 227(1–2):156–164, PMID: 16978759, <https://doi.org/10.1016/j.tox.2006.08.004>.

- ATSDR (Agency for Toxic Substances and Disease Registry). 2013. *Health Consultation, Exposure Investigation Report, Perfluorochemical Serum Sampling in the Vicinity of Decatur, Alabama: Morgan, Lawrence, and Limestone Counties*. Atlanta, GA: U.S. Department of Health and Human Services. http://www.atsdr.cdc.gov/HAC/pha/Decatur/Perfluorochemical_Serum%20Sampling.pdf [accessed 9 October 2019].
- ATSDR. 2016. *Health Consultation, Exposure Investigation, Biological Sampling of Per- and Polyfluoroalkyl Substances (PFAS) in the Vicinity of Lawrence, Morgan, and Limestone Counties, Alabama*. Atlanta, GA: U.S. Department of Health and Human Services. https://www.atsdr.cdc.gov/HAC/pha/BiologicalSampling/Biological_Sampling_of_Substances_in_Alabama_EI%20Report_11-28-2016_508.pdf [accessed 8 July 2019].
- ATSDR. 2021. Toxicological profile for perfluoroalkyls. Last updated March 2020. <https://www.atsdr.cdc.gov/ToxProfiles/tp200-p.pdf> [accessed 10 April 2020].
- Bartell SM, Calafat AM, Lyu C, Kato K, Ryan PB, Steenland K. 2010. Rate of decline in serum PFOA concentrations after granular activated carbon filtration at two public water systems in Ohio and West Virginia. *Environ Health Perspect* 118(2):222–228, PMID: 20123620, <https://doi.org/10.1289/ehp.0901252>.
- Brantsæter AL, Whitworth KW, Ydersbond TA, Haug LS, Haugen M, Knutsen HK, et al. 2013. Determinants of plasma concentrations of perfluoroalkyl substances in pregnant Norwegian women. *Environ Int* 54:74–84, PMID: 23419425, <https://doi.org/10.1016/j.envint.2012.12.014>.
- CA Water Boards (California State Water Resources Control Board). 2022. Perfluorooctanoic acid (PFOA) and Perfluorooctanesulfonic acid (PFOS). https://www.waterboards.ca.gov/drinking_water/certific/drinkingwater/PFOA_PFOS.html [accessed 22 January 2022].
- CDC (Centers for Disease Control and Prevention). 2019. *Fourth National Report on Human Exposure to Environmental Chemicals*. Atlanta, GA: U.S. Department of Health and Human Services, CDC.
- Cordner A, De La Rosa VY, Schaidler LA, Rudel RA, Richter L, Brown P. 2019. Guideline levels for PFOA and PFOS in drinking water: the role of scientific uncertainty, risk assessment decisions, and social factors. *J Expo Sci Environ Epidemiol* 29(2):157–171, PMID: 30622333, <https://doi.org/10.1038/s41370-018-0099-9>.
- Costa G, Sartori S, Consonni D. 2009. Thirty years of medical surveillance in perfluorooctanoic acid production workers. *J Occup Environ Med* 51(3):364–372, PMID: 19225424, <https://doi.org/10.1097/JOM.0b013e3181965d80>.
- Daly ER, Chan BP, Talbot EA, Nassif J, Bean C, Cavallo SJ, et al. 2018. Per- and polyfluoroalkyl substance (PFAS) exposure assessment in a community exposed to contaminated drinking water, New Hampshire, 2015. *Int J Hyg Environ Health* 221(3):569–577, PMID: 29514764, <https://doi.org/10.1016/j.ijheh.2018.02.007>.
- Dunson DB. 2001. Commentary: practical advantages of Bayesian analysis of epidemiologic data. *Am J Epidemiol* 153(12):1222–1226, PMID: 11415958, <https://doi.org/10.1093/aje/153.12.1222>.
- Eghegy PP, Lorber M. 2011. An assessment of the exposure of Americans to perfluorooctane sulfonate: a comparison of estimated intake with values inferred from NHANES data. *J Expo Sci Environ Epidemiol* 21(2):150–168, PMID: 20145679, <https://doi.org/10.1038/jes.2009.73>.
- EGLE (Michigan Department of Environment, Great Lakes, and Energy). 2020. New State Drinking Water Standards Pave Way for Expansion of Michigan's PFAS Clean-Up Efforts. <https://content.govdelivery.com/accounts/MIDEQ/bulletins/2988e74> [accessed 19 January 2022].
- Emmett EA, Shofer FS, Zhang H, Freeman D, Desai C, Shaw LM. 2006. Community exposure to perfluorooctanoate: relationships between serum concentrations and exposure sources. *J Occup Environ Med* 48(8):759–770, PMID: 16902368, <https://doi.org/10.1097/01.jom.0000232486.07658.74>.
- Evans S, Andrews D, Stoiber T, Naidenko O. 2020. PFAS Contamination of Drinking Water Far More Prevalent than Previously Reported. <https://www.ewg.org/research/national-pfas-testing/> [accessed 10 March 2020].
- EWG (Environmental Working Group). 2019a. EWG's Tap Water Database 2019 Update. <https://www.ewg.org/tapwater/> [accessed 4 December 2019].
- EWG. 2019b. Perfluorohexane Sulfonate (PFHxS): Horsham Water and Sewer Authority. <https://www.ewg.org/tapwater/system-contaminant.php?pw=PA1460033&contamcode=E204> [accessed 4 December 2019].
- FoodRisk. 2020. Food Commodity Intake Database. What We Eat in America. <https://fcid.foodrisk.org/percentiles> [accessed 10 March 2020].
- Gelman A. 2006. Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Anal* 1(3):515–534, <https://doi.org/10.1214/06-BA117A>.
- Gelman A, Carlin JB, Stern HS, Dunson DB, Vehtari A, Rubin DB. 2013. *Bayesian Data Analysis*. 3rd ed. Boca Raton, FL: Chapman and Hall/CRC Press.
- Graber JM, Alexander C, Laumbach RJ, Black K, Strickland PO, Georgopoulos PG, et al. 2019. Per- and polyfluoroalkyl substances (PFAS) blood levels after contamination of a community water supply and comparison with 2013–2014 NHANES. *J Expo Sci Environ Epidemiol* 29(2):172–182, PMID: 30482936, <https://doi.org/10.1038/s41370-018-0096-z>.
- Harada K, Inoue K, Morikawa A, Yoshinaga T, Saito N, Koizumi A. 2005. Renal clearance of perfluorooctane sulfonate and perfluorooctanoate in humans and their species-specific excretion. *Environ Res* 99(2):253–261, PMID: 16194675, <https://doi.org/10.1016/j.envres.2004.12.003>.
- Hölzer J, Göen T, Rauchfuss K, Kraft M, Angerer J, Kleeschulte P, et al. 2009. One-year follow-up of perfluorinated compounds in plasma of German residents from Arnsberg formerly exposed to PFOA-contaminated drinking water. *Int J Hyg Environ Health* 212(5):499–504, PMID: 19464951, <https://doi.org/10.1016/j.ijheh.2009.04.003>.
- Hölzer J, Midasch O, Rauchfuss K, Kraft M, Reupert R, Angerer J, et al. 2008. Biomonitoring of perfluorinated compounds in children and adults exposed to perfluorooctanoate-contaminated drinking water. *Environ Health Perspect* 116(5):651–657, PMID: 18470314, <https://doi.org/10.1289/ehp.11064>.
- Hu XC, Andrews DQ, Lindstrom AB, Bruton TA, Schaidler LA, Grandjean P, et al. 2016. Detection of poly- and perfluoroalkyl substances (PFASs) in U.S. drinking water linked to industrial sites, military fire training areas, and wastewater treatment plants. *Environ Sci Technol Lett* 3(10):344–350, PMID: 27752509, <https://doi.org/10.1021/acs.estlett.6b00260>.
- Huang MC, Dzierlenga AL, Robinson VG, Waidyanatha S, DeVito MJ, Eifrid MA, et al. 2019. Toxicokinetics of perfluorobutane sulfonate (PFBS), perfluorohexane-1-sulphonic acid (PFHxS), and perfluorooctane sulfonic acid (PFOS) in male and female Hsd:Sprague Dawley SD rats after intravenous and gavage administration. *Toxicol Rep* 6:645–655, PMID: 31334035, <https://doi.org/10.1016/j.toxrep.2019.06.016>.
- HWSA (Horsham Water and Sewer Authority). 2014. 2014 Water Quality Report: Horsham Water and Sewer Authority. https://www.horshamwater-sewer.com/sites/default/files/ccr_reports/2014%20CCR.pdf [accessed 23 December 2019].
- HWSA. 2018. 2018 Water Quality Report: Horsham Water and Sewer Authority. https://www.horshamwater-sewer.com/sites/default/files/ccr_reports/2018_ccr.pdf [accessed 25 November 2019].
- IARC (International Agency for Research on Cancer). 2017. *Some Chemicals Used as Solvents and in Polymer Manufacture*. Lyon, France: IARC.
- Johnson J, Kari A, Huest C, Williams A. 2017. Community Exposure to PFCs in Washington County, Minnesota: the East Metro Perfluorochemicals Biomonitoring Pilot Project. https://www.cleanwateraction.org/files/publications/mn/pfc_biomonitoring_study_results.pdf [accessed 4 December 2019].
- Koponen J, Winkens K, Airaksinen R, Berger U, Vestergren R, Cousins IT, et al. 2018. Longitudinal trends of per- and polyfluoroalkyl substances in children's serum. *Environ Int* 121(pt 1):591–599, PMID: 30308470, <https://doi.org/10.1016/j.envint.2018.09.006>.
- Li Y, Fletcher T, Mucs D, Scott K, Lindh CH, Tallving P, et al. 2018. Half-lives of PFOS, PFHxS and PFOA after end of exposure to contaminated drinking water. *Occup Environ Med* 75(1):46–51, PMID: 29133598, <https://doi.org/10.1136/oemed-2017-104651>.
- Lorber M, Eghegy PP. 2011. Simple intake and pharmacokinetic modeling to characterize exposure of Americans to perfluorooctanoic acid, PFOA. *Environ Sci Technol* 45(19):8006–8014, PMID: 21517063, <https://doi.org/10.1021/es103718h>.
- MDH (Minnesota Department of Health). 2021. Comparison of State Water Guidance and Federal Drinking Water Standards. <https://www.health.state.mn.us/communities/environment/risk/guidance/waterguidance.html> [accessed 19 January 2022].
- Nakagawa H, Hirata T, Terada T, Jutabha P, Miura D, Harada KH, et al. 2008. Roles of organic anion transporters in the renal excretion of perfluorooctanoic acid. *Basic Clin Pharmacol Toxicol* 103(1):1–8, PMID: 18373647, <https://doi.org/10.1111/j.1742-7843.2007.00155.x>.
- NJDEP (New Jersey Department of Environmental Protection). 2021. PFAS in Drinking Water. <https://www.nj.gov/dep/watersupply/pfas/> [accessed 19 January 2022].
- North Wales Water Authority. 2018. Water Quality Report 2018. <https://www.nwwater.com/images/documents/ccrs/2018ccrwebversion.pdf> [accessed 19 October 2019].
- NTP (National Toxicology Program). 2016. *NTP Monograph: Immunotoxicity Associated with Exposure to Perfluorooctanoic Acid or Perfluorooctane Sulfonate*. https://ntp.niehs.nih.gov/ntp/ohat/pfoa-pfos/pfoa_pfosmonograph_508.pdf [accessed 19 October 2019].
- Nuzzo R. 2015. Chance: peace talks in the probability wars. *New Scientist*, Physics Features. <https://www.newscientist.com/article/mg22530121-200-chance-peace-talks-in-the-probability-wars/> [accessed 20 October 2019].
- Olsen GW, Burris JM, Ehresman DJ, Froehlich JW, Seacat AM, Butenhoff JL, et al. 2007. Half-life of serum elimination of perfluorooctanesulfonate, perfluorohexanesulfonate, and perfluorooctanoate in retired fluorochemical production workers. *Environ Health Perspect* 115(9):1298–1305, PMID: 17805419, <https://doi.org/10.1289/ehp.10009>.
- Olsen GW, Mair DC, Lange CC, Harrington LM, Church TR, Goldberg CL, et al. 2017. Per- and polyfluoroalkyl substances (PFAS) in American Red Cross adult blood

- donors, 2000–2015. *Environ Res* 157:87–95, PMID: 28528142, <https://doi.org/10.1016/j.envres.2017.05.013>.
- Park SK, Peng Q, Ding N, Mukherjee B, Harlow SD. 2019. Determinants of per- and polyfluoroalkyl substances (PFAS) in midlife women: evidence of racial/ethnic and geographic differences in PFAS exposure. *Environ Res* 175:186–199, PMID: 31129528, <https://doi.org/10.1016/j.envres.2019.05.028>.
- Penn DOH (Pennsylvania Department of Health). 2019. *PFAS Exposure Assessment Technical Toolkit (PEAT) Pilot Project. Final Report. Division of Environmental Health Epidemiology*. <https://www.health.pa.gov/topics/Documents/Environmental%20Health/PEAT%20Pilot%20Project%20Final%20Report%20April%2029%202019.pdf> [accessed 5 June 2020].
- Post GB, Louis JB, Lippincott RL, Procopio NA. 2013. Occurrence of perfluorinated compounds in raw water from New Jersey public drinking water systems. *Environ Sci Technol* 47(23):13266–13275, PMID: 24187954, <https://doi.org/10.1021/es402884x>.
- Seals R, Bartell SM, Steenland K. 2011. Accumulation and clearance of perfluorooctanoic acid (PFOA) in current and former residents of an exposed community. *Environ Health Perspect* 119(1):119–124, PMID: 20870569, <https://doi.org/10.1289/ehp.1002346>.
- Silver N. 2012. *The Signal and the Noise: Why So Many Predictions Fail—But Some Don't*. New York, NY: Penguin Press.
- Singer AB, Whitworth KW, Haug LS, Sabaredzovic A, Impinen A, Papadopoulou E, et al. 2018. Menstrual cycle characteristics as determinants of plasma concentrations of perfluoroalkyl substances (PFASs) in the Norwegian Mother and Child Cohort (MoBa study). *Environ Res* 166:78–85, PMID: 29879567, <https://doi.org/10.1016/j.envres.2018.05.019>.
- Sunderland EM, Hu XC, Dassuncao C, Tokranov AK, Wagner CC, Allen JG. 2019. A review of the pathways of human exposure to poly- and perfluoroalkyl substances (PFASs) and present understanding of health effects. *J Expo Sci Environ Epidemiol* 29(2):131–147, PMID: 30470793, <https://doi.org/10.1038/s41370-018-0094-1>.
- Sundström M, Chang SH, Noker PE, Gorman GS, Hart JA, Ehresman DJ, et al. 2012. Comparative pharmacokinetics of perfluorohexanesulfonate (PFHxS) in rats, mice, and monkeys. *Reprod Toxicol* 33(4):441–451, PMID: 21856411, <https://doi.org/10.1016/j.reprotox.2011.07.004>.
- Thompson J, Lorber M, Toms LML, Kato K, Calafat AM, Mueller JF. 2010. Use of simple pharmacokinetic modeling to characterize exposure of Australians to perfluorooctanoic acid and perfluorooctane sulfonic acid. *Environ Int* 36(4):390–397, PMID: 20236705, <https://doi.org/10.1016/j.envint.2010.02.008>.
- U.S. EPA (U.S. Environmental Protection Agency). 2014. *Guidance for Applying Quantitative Data to Develop Data-Derived Extrapolation Factors for Interspecies and Intraspecies Extrapolation*. EPA/R-14/002F. Washington, DC: U.S. EPA, Risk Assessment Forum, Office of the Science Advisor. <https://www.epa.gov/sites/production/files/2015-01/documents/ddef-final.pdf> [accessed 18 November 2022].
- U.S. EPA. 2016. *Drinking Water Health Advisory for Perfluorooctane Sulfonate (PFOS)*. https://www.epa.gov/sites/production/files/2016-05/documents/pfos_health_advisory_final_508.pdf [accessed 28 March 2021].
- U.S. EPA. 2017. *Unregulated Contaminant Monitoring Rule 3 (UCMR 3), (2013–2015) Occurrence Data*. <https://catalog.data.gov/dataset/unregulated-contaminant-monitoring-rule-3-ucmr-3-2013-2015-occurrence-data> [accessed 18 October 2019].
- U.S. EPA. 2018. *2018 Edition of the Drinking Water Standards and Health Advisories Tables*. EPA 822-F-18-001. <https://www.epa.gov/system/files/documents/2022-01/dwtable2018.pdf> [accessed 18 November 2022].
- U.S. EPA. 2021. *PFAS Strategic Roadmap: EPA's Commitments to Action 2021–2024*. EPA-100-K-21-002. https://www.epa.gov/system/files/documents/2021-10/pfas-roadmap_final-508.pdf [accessed 9 November 2021].
- U.S. EPA. 2022. Technical Fact Sheet: Drinking Water Health Advisories for Four PFAS (PFOA, PFOS, GenX chemicals, and PFBS). EPA 822-F-22-00. <https://www.epa.gov/system/files/documents/2022-06/technical-factsheet-four-PFAS.pdf> [accessed 2 August 2022].
- Wambaugh JF, Setzer RW, Pitruzzello AM, Liu J, Reif DM, Kleinstreuer NC, et al. 2013. Dosimetric anchoring of *in vivo* and *in vitro* studies for perfluorooctanoate and perfluorooctanesulfonate. *Toxicol Sci* 136(2):308–327, PMID: 24046276, <https://doi.org/10.1093/toxsci/ktf204>.
- Warrington Township. 2014. *2014 Warrington Township Water Quality Report*. <https://www.warringtontownship.org/waterqualityreports/> [accessed 4 December 2019].
- Warrington Township. 2018. *2018 Warrington Township Water Quality Report*. <https://www.warringtontownship.org/waterqualityreports/> [accessed 22 November 2019].
- Wetmore BA, Wambaugh JF, Allen B, Ferguson SS, Sochaski MA, Setzer RW, et al. 2015. Incorporating high-throughput exposure predictions with dosimetry-adjusted *in vitro* bioactivity to inform chemical toxicity testing. *Toxicol Sci* 148(1):121–136, PMID: 26251325, <https://doi.org/10.1093/toxsci/ktv171>.
- WHO (World Health Organization), ILO (International Labour Organization), UNEP (U.N. Environment Programme). 2005. *Chemical-Specific Adjustment Factors for Interspecies Differences and Human Variability: Guidance Document for Use of Data in Dose/Concentration–Response Assessment*. IPCS Harmonization Project Document No. 2. Geneva, Switzerland: World Health Organization. <https://apps.who.int/iris/rest/bitstreams/51378/retrieve> [accessed 18 November 2022].
- Wickham H, Averick M, Bryan J, Chang W, D'Agostino McGowan L, François R, et al. 2019. Welcome to the tidyverse. *J Open Source Softw* 4(43):1686, <https://doi.org/10.21105/joss.01686>.
- WMA (Warminster Municipal Authority). 2018. *2018 Annual Drinking Water Quality Report*. PWSID No. 1090069. https://www.warminsterauthority.com/sites/default/files/ccr_reports/2018ccr.pdf [accessed 4 December 2019].
- Wong F, MacLeod M, Mueller JF, Cousins IT. 2014. Enhanced elimination of perfluorooctane sulfonic acid by menstruating women: evidence from population-based pharmacokinetic modeling. *Environ Sci Technol* 48(15):8807–8814, PMID: 24943117, <https://doi.org/10.1021/es500796y>.
- Worley RR, Moore SM, Tierney BC, Ye X, Calafat AM, Campbell S, et al. 2017a. Per- and polyfluoroalkyl substances in human serum and urine samples from a residentially exposed community. *Environ Int* 106:135–143, PMID: 28645013, <https://doi.org/10.1016/j.envint.2017.06.007>.
- Worley RR, Yang X, Fisher J. 2017b. Physiologically based pharmacokinetic modeling of human exposure to perfluorooctanoic acid suggests historical non drinking-water exposures are important for predicting current serum concentrations. *Toxicol Appl Pharmacol* 330:9–21, PMID: 28684146, <https://doi.org/10.1016/j.taap.2017.07.001>.
- Yu CH, Riker CD, Lu SE, Fan ZT. 2020. Biomonitoring of emerging contaminants, perfluoroalkyl and polyfluoroalkyl substances (PFAS), in New Jersey adults in 2016–2018. *Int J Hyg Environ Health* 223(1):34–44, PMID: 31679856, <https://doi.org/10.1016/j.ijheh.2019.10.008>.
- Zhang Y, Beesoon S, Zhu L, Martin JW. 2013. Biomonitoring of perfluoroalkyl acids in human urine and estimates of biological half-life. *Environ Sci Technol* 47(18):10619–10627, PMID: 23980546, <https://doi.org/10.1021/es401905e>.